Mathematics 40
Q10

You may choose to use the zoom meeting chat or open your mic if you have a question about the quiz.

Let $L : \mathbb{R}^3 \to P_2$ be the linear transformation (no need to verify) defined by

$$L((a,b,c)) = \left(2a - \frac{b}{2} - \frac{c}{2}\right)x^2 + (6b+c)x + \left(a + \frac{3b}{2} - \frac{3c}{2}\right).$$

Let $B_1 = \left\{\left(\frac{1}{2},0,0\right),(1,0,2),(1,1,-1)\right\}$, $B_2 = \{x^2+1, 2x-3, 5x+4\}$, $B_3 = \{x^2+1, x^2+2x-2, 7x+1\}$, $B_4 = \{e_1, e_2, e_3\}$.

1. Compute for $[L]_{B_1}^{B_2}$
2. Find nonsingular matrices P and Q such that $P[L]_{B_1}^{B_3}Q = [L]_{B_4}^{B_3}$.
3. Solve for $[(3,0,2)]_{B_1}$. Using this and item 1, solve for $[L(3,0,2)]_{B_2}$.

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1. $L((1/2,0,0)) = (2\cdot 1/2 - 0 - 0)x^2 + (0+0)x + (\frac{1}{2} + 0 - 0) = x^2 + 1/2$

$= a_1(x^2+1) + a_2(2x-3) + a_3(5x+4)$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 5 & 0 \\ 1 & -3 & 4 & 1/2 \end{bmatrix} \xrightarrow{-r_1+r_3 \to r_3} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 5 & 0 \\ 0 & -3 & 4 & -1/2 \end{bmatrix} \xrightarrow[-r_3 \to r_3]{r_2+r_3 \to r_3} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 5 & 0 \\ 0 & 1 & -9 & 1/2 \end{bmatrix} \xrightarrow[r_2 \leftrightarrow r_3]{-2r_3+r_2 \to r_2} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -9 & 1/2 \\ 0 & 0 & 23 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -9 & 1/2 \\ 0 & 0 & 1 & -1/23 \end{bmatrix}$$

$a_3 = -1/23$

$a_2 = 1/2 + 9a_3 = 1/2 - 9/23 = \frac{23-18}{46} = \frac{5}{46}$

$a_1 = 1$

$L((1,0,2)) = (2 - 0 - 2/2)x^2 + (0+2)x + (1+0-2) = x^2 + 2x - 2$

$= a_1(x^2+1) + a_2(2x-3) + a_3(5x+4)$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 5 & 2 \\ 1 & -3 & 4 & -2 \end{bmatrix} \xrightarrow{-r_1+r_3 \to r_3} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 5 & 2 \\ 0 & -3 & 4 & -3 \end{bmatrix} \xrightarrow{\frac{1}{2}r_2 \to r_2} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 5/2 & 1 \\ 0 & -3 & 4 & -3 \end{bmatrix} \xrightarrow{3r_2+r_3 \to r_3} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 5/2 & 1 \\ 0 & 0 & 23/2 & 0 \end{bmatrix}$$

$23/2\, a_3 = 0$

$\Rightarrow a_3 = 0$

$a_2 = 1 - 5/2\, a_3 = 1$

$a_1 = 1$

$L((1,1,-0)) = (2 - 1/2 + 1/2)x^2 + (6-1)x + (1 + 3/2 + 3/2) = 2x^2 + 5x + 4$

$= a_1(x^2+1) + a_2(2x-3) + a_3(5x+4)$

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 2 & 5 & 5 \\ 1 & -3 & 4 & 4 \end{bmatrix} \xrightarrow{-r_1+r_3 \to r_3} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 2 & 5 & 5 \\ 0 & -3 & 4 & 2 \end{bmatrix} \xrightarrow[\substack{-r_3 \to r_3 \\ r_3 \leftrightarrow r_2}]{r_2+r_3 \to r_3} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & -9 & -7 \\ 0 & 2 & 5 & 5 \end{bmatrix} \xrightarrow{-2r_2+r_3 \to r_3} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & -9 & -7 \\ 0 & 0 & 23 & 19 \end{bmatrix} \xrightarrow{\frac{1}{23}r_3 \to r_3} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & -9 & -7 \\ 0 & 0 & 1 & 19/23 \end{bmatrix}$$

$a_3 = 19/23$

$a_2 = -7 + 9a_3 = -7 + 9(19/23)$

$= -7 + 171/23$

$= 10/23$

$a_1 = 2$

$$\boxed{\therefore [L]_{B_1}^{B_2} = \begin{bmatrix} 1 & 1 & 2 \\ 5/46 & 1 & 10/23 \\ -1/23 & 0 & 19/23 \end{bmatrix}}$$

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2. NOTE: $[P]_{B_2}^{B_3}[P]_{B_1}^{B_2} = [P]_{B_1}^{B_3}$

Let: $P = [L]_{B_2}^{B_3}$, $Q = [L]_{B_4}^{B_1}$

$\Rightarrow P[L]_{B_1}^{B_2}Q = [L]_{B_4}^{B_3} = [L]_{B_2}^{B_3}[L]_{B_1}^{B_2}Q = [L]_{B_4}^{B_3} \Rightarrow [L]_{B_1}^{B_3}[L]_{B_4}^{B_1} = [L]_{B_4}^{B_3} \Rightarrow [L]_{B_4}^{B_3} \overset{\checkmark}{=} [L]_{B_4}^{B_3}$

- Finding $[L]_{B_2}^{B_3}$

$x^2+1 = a_1(x^2+1) + a_2(x^2+2x-2) + a_3(7x+1)$

$\Rightarrow a_1 = 1, a_2 = 0, a_3 = 0$

$2x-3 = a_1(x^2+1) + a_2(x^2+2x-2) + a_3(7x+1)$

gives the augmented matrix

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 7 & 2 \\ 1 & -2 & 1 & -3 \end{bmatrix} \xrightarrow{-r_1+r_3\to r_3} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 7 & 2 \\ 0 & -3 & 1 & -3 \end{bmatrix} \xrightarrow{\frac{1}{2}r_2\to r_2} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 7/2 & 1 \\ 0 & -3 & 1 & -3 \end{bmatrix} \xrightarrow{3r_2+r_3\to r_3} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 7/2 & 1 \\ 0 & 0 & 23/2 & 0 \end{bmatrix}$$

$23/2\, a_3 = 0 \Rightarrow a_3 = 0$

$a_2 = 1 - 7/2\, a_3 = 1$

$a_1 = -a_2 \Rightarrow a_1 = -1$

$5x+4 = a_1(x^2+1) + a_2(x^2+2x-2) + a_3(7x+1)$

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 7 & 5 \\ 1 & -2 & 1 & 4 \end{bmatrix} \xrightarrow{-r_1+r_3\to r_3} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 2 & 7 & 5 \\ 0 & -3 & 1 & 4 \end{bmatrix} \xrightarrow{\frac{1}{2}r_2\to r_2} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 7/2 & 5/2 \\ 0 & -3 & 1 & 4 \end{bmatrix} \xrightarrow{3r_2+r_3\to r_3} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 7/2 & 5/2 \\ 0 & 0 & 23/2 & 25/2 \end{bmatrix} \xrightarrow{2/23\, r_3\to r_3} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 7/2 & 5/2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{-7/2\, r_3+r_2\to r_2}$$

$$= \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{-r_2+r_1\to r_1} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$a_1 = 1, a_2 = -1, a_3 = 1$

$$\therefore P = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

- Finding $[L]_{B_4}^{B_1}$

$(1,0,0) = a_1(1/2, 0, 0) + a_2(1,0,2) + a_3(1,1,-1)$

$= a_1 = 2, a_2 = 0, a_3 = 0$

$(0,1,0) = a_1(1/2, 0, 0) + a_2(1,0,2) + a_3(1,1,-1)$

$$\begin{bmatrix} 1/2 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 2 & -1 & 0 \end{bmatrix} \xrightarrow{2r_1\to r_1} = \begin{bmatrix} 1 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 2 & -1 & 0 \end{bmatrix} \xrightarrow[\frac{1}{2}r_2\to r_2]{r_2\leftrightarrow r_3} = \begin{bmatrix} 1 & 2 & 2 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow[-2r_3+r_1\to r_1]{1/2\, r_3+r_2\to r_2} = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{-2r_2+r_1\to r_1} = \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$a_1 = -3$

$a_2 = 1/2$

$a_3 = 1$

$(0,0,1) = a_1(1/2, 0, 0) + a_2(1,0,2) + a_3(1,1,-1)$

$$\begin{bmatrix} 1/2 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & -1 & 1 \end{bmatrix} \xrightarrow{2r_1\to r_1} = \begin{bmatrix} 1 & 2 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & -1 & 1 \end{bmatrix} \xrightarrow[\frac{1}{2}r_2\to r_2]{r_2\leftrightarrow r_3} = \begin{bmatrix} 1 & 2 & 2 & 0 \\ 0 & 1 & -1/2 & 1/2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[-2r_3+r_1\to r_1]{1/2\, r_3+r_2\to r_2} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{-2r_2+r_1\to r_1} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$a_1 = -1$

$a_2 = 1/2$

$a_3 = 0$

$$\therefore Q = \begin{bmatrix} 2 & -3 & -1 \\ 0 & 1/2 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

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3. Solve for $[(3,0,2)]_{B_1}$

$(3,0,2) = a_1(1/2, 0, 0) + a_2(1,0,2) + a_3(1,1,-1)$

augmented matrix

$$\begin{bmatrix} \frac{1}{2} & 1 & 1 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & -1 & 2 \end{bmatrix} \xrightarrow{2r_1\to r_1} \xrightarrow{r_2\leftrightarrow r_3} = \begin{bmatrix} 1 & 2 & 2 & 6 \\ 0 & 2 & -1 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{1/2\, r_2\to r_2} = \begin{bmatrix} 1 & 2 & 2 & 6 \\ 0 & 1 & -1/2 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[-2r_3+r_1\to r_1]{1/2\, r_3+r_2\to r_2} = \begin{bmatrix} 1 & 2 & 0 & 6 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{-2r_2+r_1\to r_1} = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$a_1 = 4$

$a_2 = 1$

$a_3 = 0$

$$\therefore [(3,0,2)]_{B_1} = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$$

Solve for $[L(3,0,2)]_{B_2} = [L]_{B_1}^{B_2} [(3,0,2)]_{B_1}$

$$= \begin{bmatrix} 1 & 1 & 2 \\ 5/46 & 1 & 10/23 \\ -1/23 & 0 & 14/23 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} (4+1+0) \\ (20/46+1+0) \\ (-4/23+0+0) \end{bmatrix} = \begin{bmatrix} 4 \\ 10/23+1 \\ -4/23 \end{bmatrix} = \begin{bmatrix} 4 \\ 33/23 \\ -4/23 \end{bmatrix}$$

$$\therefore [L(3,0,2)]_{B_2} = \begin{bmatrix} 4 \\ 33/23 \\ -4/23 \end{bmatrix}$$